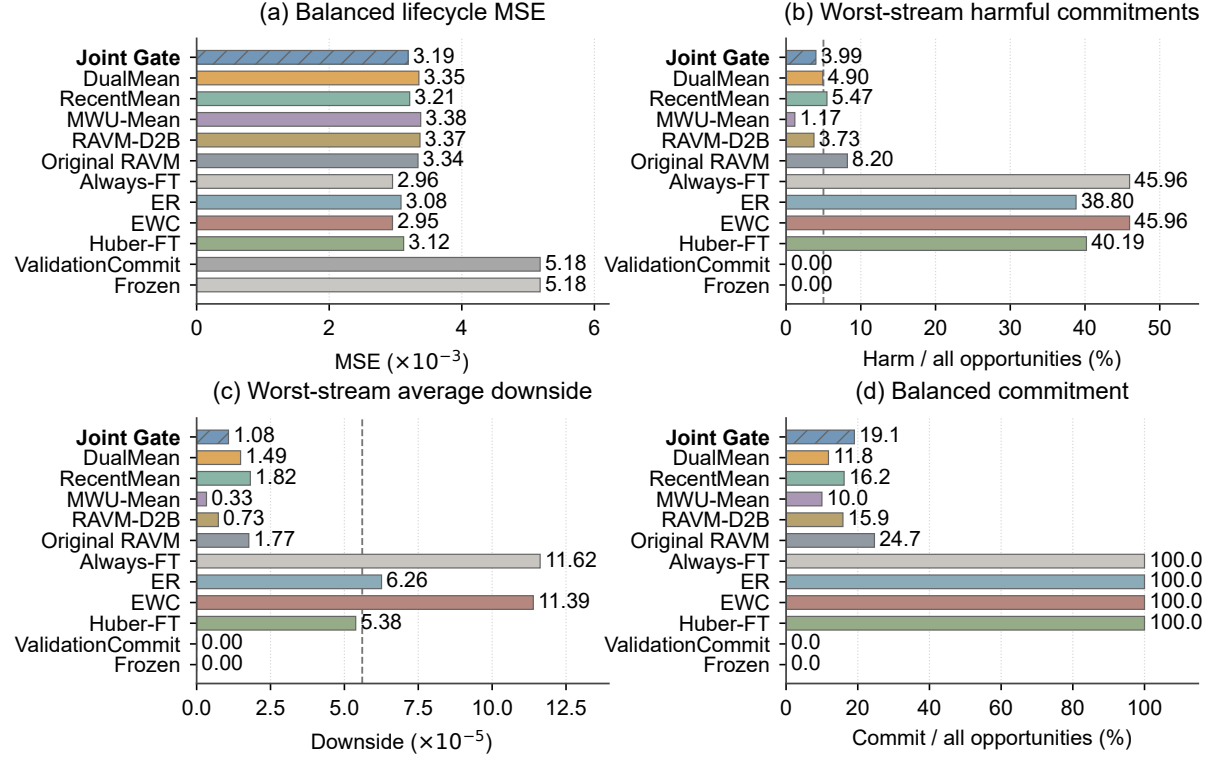
(a) Balanced lifecycle MSE
Joint Gate
DualMean
RecentMean
MWU-Mean
RAVM-D2B
Original RAVM
Always-FT
ER
EWC
Huber-FT
ValidationCommit
Frozen
3.19
3.35
3.21
3.38
3.37
3.34
2.96
3.08
2.95
3.12
5.18
5.18
0
2
4
6
MSE (×10⁻³)
(b) Worst-stream harmful commitments
3.99
4.90
5.47
1.17
3.73
8.20
45.96
38.80
45.96
40.19
0.00
0.00
0
10
20
30
40
50
Harm / all opportunities (%)
(c) Worst-stream average downside
1.08
1.49
1.82
0.33
0.73
1.77
11.62
6.26
11.39
5.38
0.00
0.00
0.0
2.5
5.0
7.5
10.0
12.5
Downside (×10⁻⁵)
(d) Balanced commitment
19.1
11.8
16.2
10.0
15.9
24.7
100.0
100.0
100.0
100.0
0.0
0.0
0
20
40
60
80
100
Commit / all opportunities (%)